

Problem

Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1.

Input Format

The first line contains an integer T , the number of test cases. For each test case: The first line contains two integers V and N . V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

$1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1.

Sample Input 0

7 2
2 4

```

21         for(int i=1;i<=v;i++) {
22             for(int j=0;j<n;j++) {
23                 if(i>=coins[j]) {
24                     if(dp[i-coins[j]] + 1<d
25                         dp[i] = dp[i-coins[
26                             }
27                 }
28             }
29         }
30         if(dp[v]>v) {
31             System.out.println(-1);
32         }else {
33             System.out.println(dp[v]);
34         }
35     }
36 }
37 }
```

Line: 37 Col: 2



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

